

Reviewer Pack — Minimal Rank of Algebraic K-Theory over Boolean Function Com...

Entry ebe571bd094a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-27 10:07:59 UTC

Minimal Rank of Algebraic K-Theory over Boolean Function Complexity

Entry ID: ebe571bd094a **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 10:07:59 UTC

1. Conjecture

Field A (mathematical branch): Algebraic K-theory **Field B** (complexity object): Boolean Function Complexity

Statement:

[‘For any boolean function f with n variables, the minimal rank of its algebraic K-theory group is at least $\Omega(n/\log n)$.’, ‘This implies that there exists a polynomial-time computable map from the input space to the algebraic K-theory group of f such that for any input x , the rank of the corresponding element in the group is greater than or equal to $\Omega(\log(|f(x)|))$.’, ‘Moreover, this map satisfies the natural group action on the boolean function complexity.’]

Rationale (proposer’s reasoning):

[‘Algebraic K-theory provides a rich algebraic structure that has been little explored in complexity theory. If an invariant of boolean functions could be computed using algebraic K-theory, it would give new insights into understanding the complexity of these functions.’, ‘The use of algebraic K-theory as an invariant for boolean functions has not been extensively studied yet, and if there is a relationship between the rank of its group and the complexity of the function, this could provide a novel way to prove lower bounds on computational complexity.’]

Taxonomy category: AC0_PARITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 9a92b53fa8b70626

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, across at least 1000 randomly generated boolean functions with n variables, the mean minimal rank of the algebraic K-theory group is greater than or equal to $\Omega(n/\log n)$ and no seed has a minimal rank less than $\Omega(n/\log n)$. The conjecture is falsified if any seed produces a minimal rank less than $\Omega(n/\log n)$, or the mean minimal rank across all seeds is less than $\Omega(n/\log n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'algebraic K-theory' AND 'boolean function complexity' AND 'minimal rank' - 'K-theory' IN ('algebraic' OR 'number theory') AND 'boolean functions' AND 'complexity' - 'polynomial-time computable map' AND 'algebraic K-theory group' AND 'boolean function complexity'

Top relevant hits considered: - [http://arxiv.org/abs/2103.15474v4] Hermitian K-theory via oriented Gorenstein algebras - [http://arxiv.org/abs/2011.11755v3] Boolean algebras, Morita invariance, and the algebraic K-theory of Lawvere theories - [http://arxiv.org/abs/2410.04531v1] On the K-theory of magnetic algebras: Iwatsuka case - [http://arxiv.org/abs/1311.1421v3] Multiplicative differential algebraic K-theory and applications - [http://arxiv.org/abs/1811.09564v3] Dévissage for Waldhausen K-theory - [http://arxiv.org/abs/1710.00331v2] Hecke modules for arithmetic groups via bivariant K-theory - [http://arxiv.org/abs/2104.04021v2] A universal characterization of noncommutative motives and secondary algebraic K-theory - [s2:10.1007/10719839] LATIN 2000: Theoretical Informatics: 4th Latin American Symposium, Punta del Este, Uruguay, April 10-14, 2000 Proceeding

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=124, elapsed=240.3s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def generate_boolean_function(n):
    return [random.choice([0, 1]) for _ in range(2**n)]

def algebraic_k_theory_group(f):
    n = int(math.log2(len(f)))
    if len(f) != 2**n:
        raise ValueError("Input must be a boolean function with 2^n values")

    # Simplified version of computing the algebraic K-theory group
    # This is a placeholder and should be replaced with actual computation
    return [f.count(1), f.count(0)]

def run_trial(seed: int) -> dict:
    random.seed(seed)
```

```

n_values = [5, 10, 15, 20, 30, 40]
results = []

for n in n_values:
    f = generate_boolean_function(n)
    group = algebraic_k_theory_group(f)
    rank = max(group)

    result = {
        "metric_name": "minimal_rank",
        "metric_value": rank,
        "instances_tested": 1,
        "conjecture_holds": rank >= n / math.log(n),
        "counterexample": "" if rank >= n / math.log(n) else f"n={n}, rank={rank}"
    }
    results.append(result)

mean_rank = sum(r["metric_value"] for r in results) / len(results)
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

return {
    "seed": seed,
    "mean_rank": mean_rank,
    "support_fraction": support_fraction,
    "results": results
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")

        if "support_fraction" not in result:
            continue

        mean_rank = result["mean_rank"]
        support_fraction = result["support_fraction"]

        if support_fraction >= 0.8:
            print(f"RESULT: SUPPORTED mean={mean_rank} std=NA support_fraction={support_fraction}")
        elif any(r["conjecture_holds"] is False for r in result["results"]):
            first_failing_seed = next(s for s, r in zip(seeds, result["results"]) if not r["conjecture_holds"])
            print(f"RESULT: FALSIFIED counterexample='n={r[0][\"instances_tested\"]} rank<{r[0][\"metric_value\"]}'")
        else:
            print("RESULT: INCONCLUSIVE reason=insufficient_data")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot verify the conjecture's support or falsification. | next: Run the test again with increased time limits to ensure it completes and produces results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12642
2	preregistration	ollama_remote	glm4:latest	0	0	6167
3	novelty	ollama_remote	glm4:latest	0	0	5048
4	novelty	ollama_remote	glm4:latest	0	0	5918
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18516
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8831
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7204
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7389
9	judge	ollama_remote	glm4:latest	0	0	15187

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 86901 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/ebe571bd094a.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/ebe571bd094a.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/ebe571bd094a.tar.gz (if generated)